

4.1

$$a. SSTO = \sum_{i=1}^{20} (Y_i - \bar{Y})^2 = \sum Y_i^2 - n\bar{Y}^2$$

$$= 7825.94 - 20 \times 19.21^2$$

$$= 445.458$$

$$R^2 = \frac{SSR}{SSTO} = \frac{395.49}{445.458} = 0.8879$$

$$b. \hat{\sigma}^2 = MSE = \frac{SSE}{N-2} = \frac{151.6489}{18} = 8.4249$$

$$0.9911 = 1 - \frac{SSE}{7825.94} \Rightarrow SSE = 151.6489$$

c.

$$SSTO = 631.63$$

$$\sum \hat{e}_i^2 = SSE = 182.85$$

$$R^2 = 1 - \frac{182.85}{631.63} = 0.7105$$

4.3

$$a. \hat{Y} = 1.2 + 0.8X$$

$$\hat{Y}_0 = 1.2 + 0.8 \times 4 = 4.4$$

$$b. \hat{\sigma}^2 = \frac{SSE}{n-2} = \frac{3.6}{3} = 1.2$$

$$\bar{X} = 1$$

$$\widehat{Var}(f) = \hat{\sigma}^2 \left[1 + \frac{1}{n} + \frac{(X_0 - \bar{X})^2}{S_{XX}} \right] = 1.2 \left(1 + \frac{1}{5} + \frac{9}{10} \right) = 2.52$$

c.

$X_0 = 4$ 时 Y 的 95% prediction interval

$$4.4 \pm t_{0.025}(3) SE(f) = 4.4 \pm 3.182 \times \sqrt{2.52}$$

$$\Rightarrow [-0.6514, 9.4514]$$

$\nearrow SE(f)$ 在 $X = \bar{X}$ 时最小, prediction interval 最小

d.

$X_0 = 1$ 时, Y 的 95% prediction interval

$$4.4 \pm t_{0.025}(3) SE(f) = 4.4 \pm 3.182 \times \sqrt{1.44}$$

$$\Rightarrow [-4.8926, 13.6926]$$

e.

$$\widehat{Var}(f|X_0=1) = 1.2 \times (1 + 0.2) = 1.44$$

$X_0 = 1$ 时, Y 的 95% prediction interval

$$4.4 \pm 3.182 \times \sqrt{1.44}$$

$$\Rightarrow [-1.8184, 5.8184]$$

9.25

Call:
lm(formula = log(price) ~ sqft, data = collegetown)

Residuals:

	Min	1Q	Median	3Q	Max
	-1.32528	-0.19199	0.00122	0.21678	0.86609

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.393866	0.043288	101.50	<2e-16 ***
sqft	0.036044	0.001486	24.26	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.34 on 498 degrees of freedom
Multiple R-squared: 0.5417, Adjusted R-squared: 0.5408
F-statistic: 588.7 on 1 and 498 DF, p-value: < 2.2e-16

Slope at mean: 9.019661
Elasticity at mean: 0.9833701

Call:
lm(formula = log(price) ~ log(sqft), data = collegetown)

Residuals:

	Min	1Q	Median	3Q	Max
	-1.36864	-0.20849	0.00487	0.23204	1.21099

Coefficients:

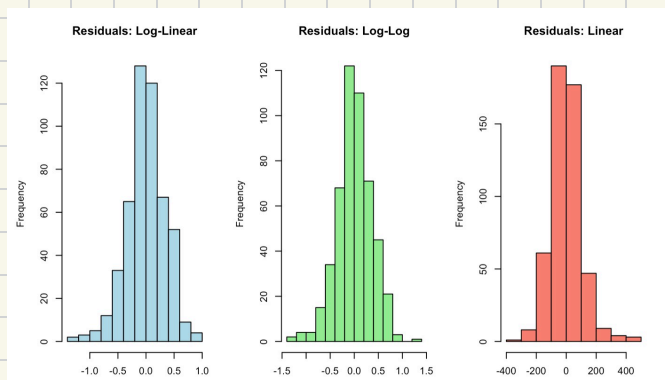
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.04965	0.15797	12.97	<2e-16 ***
log(sqft)	1.02483	0.04839	21.18	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

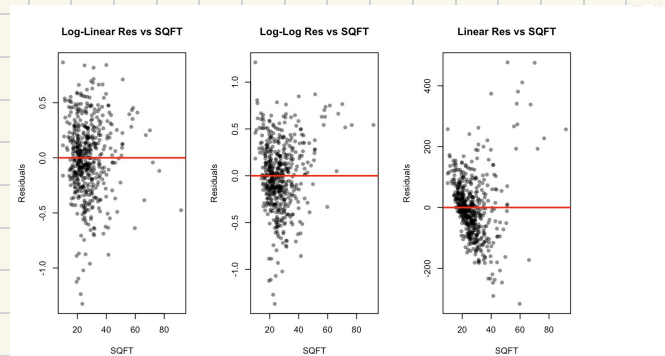
Residual standard error: 0.3643 on 498 degrees of freedom
Multiple R-squared: 0.4738, Adjusted R-squared: 0.4728
F-statistic: 448.5 on 1 and 498 DF, p-value: < 2.2e-16

Slope at mean: 9.399923
Elasticity: 1.024828

Model	R_Squared
Linear	0.6413
Log-Linear (Gen R^2)	0.6622
Log-Log (Gen R^2)	0.6445



Model	X_squared	p_value
Log-Linear	26.67899	0.000002
Log-Log	14.27871	0.000793
Linear	221.11152	0.000000



Predicted Price & 95%
PI (in thousands of
dollars)

	fit	lwr	upr
1	214.23	109.77	418.12
1	227.54	111.14	465.84
1	246.46	44.28	448.63

h' log-log / log-linear

3.5

$$a. \text{Var}(b_2|X) = \frac{\sigma^2}{\sum_{t=1}^T (x_t - \bar{x})^2}$$

$$\begin{aligned} \sum_{t=1}^T (t - \bar{t})^2 &= \sum_{t=1}^T t^2 - \frac{(\sum t)^2}{T} \\ &= \frac{T(T+1)(2T+1)}{6} - \frac{\left[\frac{T(T+1)}{2}\right]^2}{T} \\ &= \frac{T(T^2-1)}{12} \end{aligned}$$

$$\lim_{T \rightarrow \infty} \text{Var}(b_2|X) = \frac{12\sigma^2}{T(T^2-1)} = 0, \text{ 满足一致性}$$

b.

$$y_t = \beta_1 + \beta_2 (0.5)^t + e_t$$

$$\sum x_t = \frac{0.5(1-0.5^T)}{1-0.5} = 1-0.5^T$$

$$\sum x_t^2 = \sum (0.25)^t = \frac{0.25(1-0.25^T)}{1-0.25} = \frac{1}{3}(1-0.25^T)$$

$$\text{Var}(b_2|X) = \frac{\sigma^2}{\frac{1}{3}(1-0.25^T) - \frac{(1-0.5^T)^2}{T}}$$

$$\lim_{T \rightarrow \infty} \text{Var}(b_2|X) = \frac{1}{3} \neq 0, \text{ 不满足一致性}$$

c.

model a β_{10}^T , β_{10}^T 是 β_{10} 的估计量

model b β_{10}^T 是 β_{10} 的估计量

3.6

$$a. H_0: \beta_2 = 0$$

$$H_a: \beta_2 \neq 0$$

$$\alpha = 0.05$$

$$\text{test statistic} = t = \frac{b_2 - 0}{SE(b_2)} \stackrel{H_0}{\sim} t(60)$$

$$RR = \{ |t| \geq t_{0.025}(60) = 2 \}$$

$$t^* = \frac{3}{2} = 1.5 \notin RR, \text{ do not reject } H_0$$

b,

$$H_0: \beta_1 + 2\beta_2 = 5$$

$$H_a: \beta_1 + 2\beta_2 \neq 5$$

$$\alpha = 0.05$$

$$\text{test statistic} = T = \frac{b_1 + 2b_2 - 5}{SE(b_1 + 2b_2)} \stackrel{H_0}{\sim} t(b_0)$$

$$SE(b_1 + 2b_2) = \sqrt{\widehat{Var}(b_1) + 4\widehat{Var}(b_2) + 4\widehat{Cov}(b_1, b_2)} = \sqrt{3 + 16 - 8} = \sqrt{11}$$

$$RR = \{ |T| \geq t_{0.025}(b_0) = 2 \}$$

$$T^* = \frac{8-5}{\sqrt{11}} = 0.905 \notin RR, \text{ do not reject } H_0$$

c, $H_0: \beta_1 - \beta_2 + \beta_3 = 4$

$$H_a: \beta_1 - \beta_2 + \beta_3 \neq 4$$

$$\alpha = 0.05$$

$$\text{test statistic} = T = \frac{b_1 - b_2 + b_3 - 4}{SE(b_1 - b_2 + b_3)} \stackrel{H_0}{\sim} t(b_0)$$

$$SE(b_1 - b_2 + b_3) = \sqrt{\widehat{Var}(b_1) + \widehat{Var}(b_2) + \widehat{Var}(b_3) - 2\widehat{Cov}(b_1, b_2) + 2\widehat{Cov}(b_1, b_3) - 2\widehat{Cov}(b_2, b_3)}$$

$$= 4$$

$$RR = \{ |T| \geq t_{0.025}(b_0) = 2 \}$$

$$T^* = \frac{-2-4}{4} = -1.5 \notin RR, \text{ do not reject } H_0$$

5, 20

a,

$$\widehat{\beta}_{2, \text{mean}} = \frac{(A_1 + \beta_2 \bar{x}_2 + \bar{e}_2) - (A_1 + \beta_2 \bar{x}_1 + \bar{e}_1)}{\bar{x}_2 - \bar{x}_1} = \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}$$

$$E(\widehat{\beta}_{2, \text{mean}} | X) = \beta_2, \text{ unbiased}$$

b,

$$\text{Var}(\widehat{\beta}_{2, \text{mean}} | X) = \frac{1}{(\bar{x}_2 - \bar{x}_1)^2} \text{Var}(\bar{e}_2 - \bar{e}_1 | X) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$$

c,

$$\text{Var}(\widehat{\beta}_2) = E[\text{Var}(\widehat{\beta}_2 | X)] + \text{Var}(E(\widehat{\beta}_2 | X))$$

$$= \frac{4\sigma^2}{N} E\left(\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right)$$

d, $N \rightarrow \infty$ 时, $\bar{x}_2 \neq \bar{x}_1$, 则 $\lim_{N \rightarrow \infty} \text{Var}(\widehat{\beta}_2) \rightarrow 0$, 具一致性

e.

N	var_b2_OLS	var_beta_mean	Est_E_sx2_inv	Est_E_diff_inv
100	1.2436	1.6632	0.1244	0.1663
500	0.2440	0.3260	0.1220	0.1630
1000	0.1229	0.1656	0.1229	0.1656
5000	0.0241	0.0323	0.1207	0.1616

f.

$$\text{Var}(\hat{\beta}_{OLS}) < \text{Var}(\hat{\beta}_{\text{mean}})$$

且可看出 $\hat{\beta}_{\text{mean}}$ 具一致性 ($\text{Var}(\hat{\beta}_2)$ 随 n 增大而减小)

g.

$$X \sim U(0, 10)$$

$$\text{Var}(X) = \frac{(10-0)^2}{12}$$

$$s_x^2 \rightarrow \frac{100}{12}, \quad E(s_x^2)^{-1} \rightarrow \frac{12}{100} = 0.12$$

$$\bar{x}_1 \rightarrow 2.5, \quad \bar{x}_2 \rightarrow 7.5, \quad (\bar{x}_2 - \bar{x}_1) \rightarrow 5, \quad (\bar{x}_2 - \bar{x}_1)^2 \rightarrow 25$$

$$E\left(\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right) \rightarrow 0.16$$

g. T_0 没有排序.

$$\bar{x}_1 \rightarrow 5, \quad \bar{x}_2 \rightarrow 5, \quad \lim_{n \rightarrow \infty} \text{Var}(\hat{\beta}_{\text{mean}}) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} \neq 0, \quad \text{不具一致性}$$